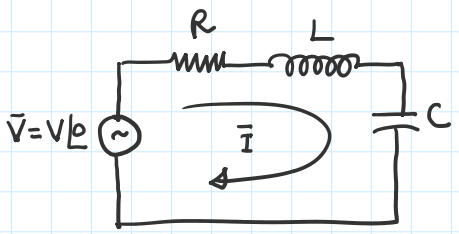
RLC- Series Circuit

$$\left.\begin{aligned} v(t) &= \sqrt{2}\,V \sin \omega t \\ &= V_m \sin \omega t \\ \bar{V} &= V \angle 0 \end{aligned}\right\}$$



KVL:

$$\bar{I}R + \bar{I}(jX_L) + \bar{I}(-jX_C) = \bar{V}$$

$$\bar{V}_R + \bar{V}_L + \bar{V}_C = \bar{V}$$

$$\bar{I}\{\underbrace{R + j(X_L - X_C)}_{Z}\} = \bar{V}$$

$$Z = Z_1 + Z_2 + Z_3 = R + j\omega L + \left(-\frac{j}{\omega C}\right) = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\bar{I} = \frac{\bar{V}}{Z} = \frac{V\angle 0}{R + j\left(\omega L - \frac{1}{\omega C}\right)} = \frac{V\angle 0}{\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]^{1/2} \angle \tan^{-1}\frac{(\omega L - 1/\omega C)}{R}}$$

$$\bar{I} = \frac{V}{|Z|} \angle -\tan^{-1}\left\{\frac{(\omega L - \frac{1}{\omega C})}{R}\right\} \quad ; \quad Z = \left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]^{1/2} ;$$

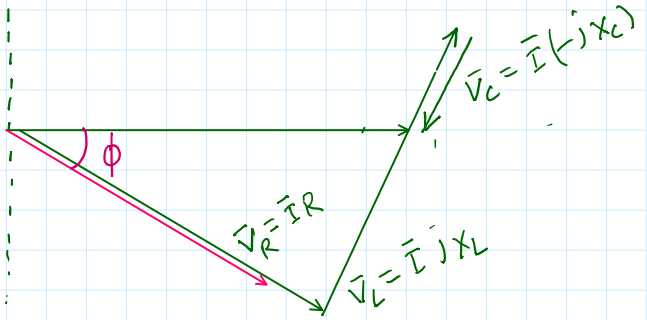
$$\downarrow$$

$$i(t) = \frac{\sqrt{2}\,V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \sin\left[\omega t - \tan^{-1}\frac{(\omega L - 1/\omega C)}{R}\right]$$

Case 1: $\omega L > \frac{1}{\omega C}$; $X_L > X_C$, ?

Z is inductive, $Z = R + j(\omega L - \frac{1}{\omega C}) = R + jX_L'$

The current will lag the voltage.

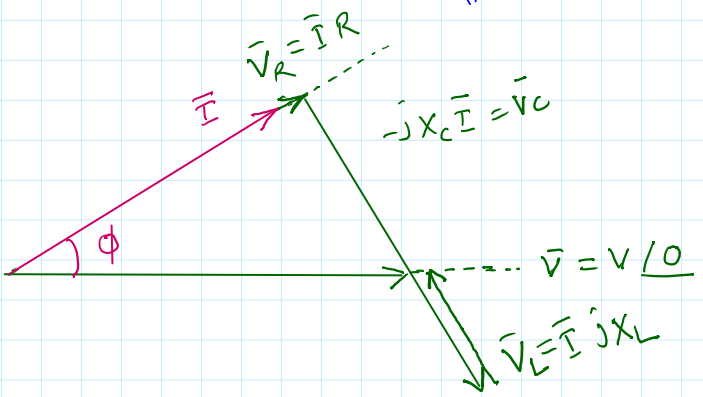


Case 2: $\omega L < \frac{1}{\omega C}$; $X_L < X_C$

Z is capacitive; $Z = R + j(X_L - X_C) = R - jX_C'$

The current will lead the voltage.

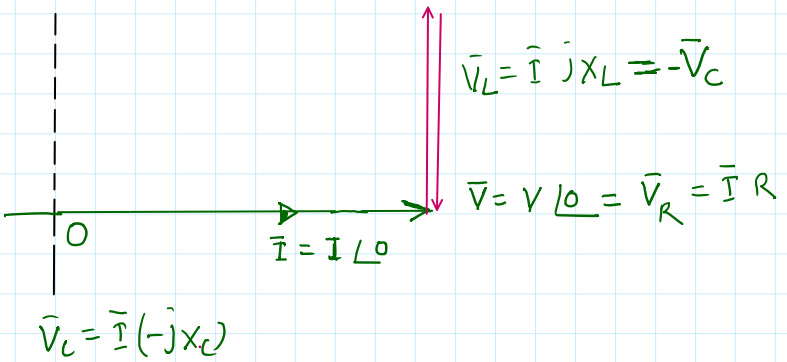
$\bar{V}_R = \bar{I}R$ ...



Case 3 : when $\omega L = \frac{1}{\omega C} \Rightarrow X_L = X_C$

Z is resistive; $Z = R$

The current will be in phase with voltage.



Condition ③ implies, $\omega L = \frac{1}{\omega C}$

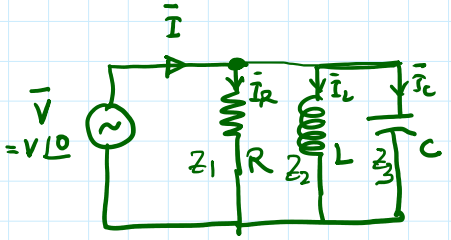
$$\omega^2 = \frac{1}{LC} \Rightarrow \omega = \omega_n = \sqrt{\frac{1}{LC}} \quad \text{rad/sec}$$

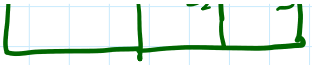
This is known as "Resonance" phenomenon.

In general, the current in time domain is expressed as

$$i(t) = \frac{\sqrt{2}\,V}{\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]^{1/2}} \sin\left\{\omega t - \tan^{-1}\frac{(\omega L - 1/\omega C)}{R}\right\}$$

RLC Parallel Circuit





$$\bar{I}_1 = \frac{\bar{V}}{z_1} \qquad z_1 = R$$

$$\bar{I}_2 = \frac{\bar{V}}{z_2} \qquad z_2 = j\omega L = jX_L$$

$$\bar{I}_3 = \frac{\bar{V}}{z_3} \qquad z_3 = \frac{-j}{\omega C} = -jX_C$$

$$\bar{I} = \bar{I}_1 + \bar{I}_2 + \bar{I}_3 = \bar{V}\left[\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right] = \bar{V}\cdot\left(\frac{1}{z_{eq}}\right)$$

$$\frac{1}{z_{eq}} = \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}$$

$$\bar{I} = \bar{V}\cdot\frac{1}{z_{eq}} = \bar{V}\cdot Y \;;\; Y = \text{Admittance} = G + jB$$

$$\bar{I} = V\angle 0\left[\frac{1}{R} + \frac{1}{j\omega L} + j\omega C\right] = V\angle 0\left[\frac{1}{R} + j\underbrace{\left(\omega C - \frac{1}{\omega L}\right)}_{G + jB}\right]$$

$$Y$$

$$G = \frac{1}{R}$$

$$B = \left(\omega C - \frac{1}{\omega L}\right)$$

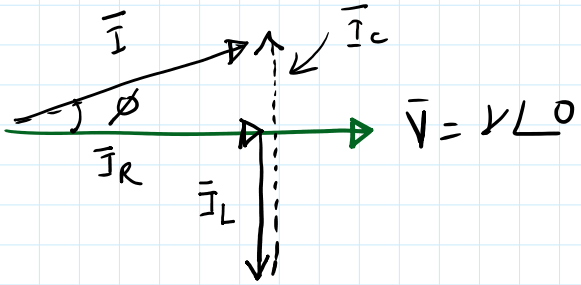
$$\omega C = \frac{1}{\omega L} \Rightarrow \omega_n = \frac{1}{\sqrt{LC}}$$

$$\bar{I} = V\left[\frac{1}{R} + j\underbrace{\left(\omega C - \frac{1}{\omega L}\right)}\right]$$

$\omega > \omega_n$; (capacitive)

$\omega < \omega_n$ (inductive)

$\omega = \omega_n$ (Resistive)



Example:

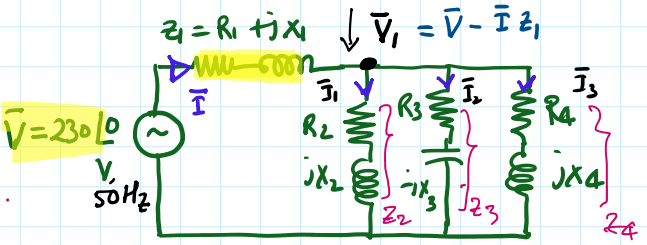
$R_1 = 10\Omega$, $L_1 = 31.8\,mH$

$R_2 = 15\Omega$, $L_2 = 63.6\,mH$

$R_3 = 3\Omega$, $C_3 = 795\,\mu F$

$R_4 = 8\Omega$, $L_4 = 19.09\,mH$

$\bar{V} = 230\angle 0\ V$ at $f = 50\,Hz$



$$z_1 = R_1 + j\omega L_1 = 10 + j2\pi \times 50 \times 31.8\times 10^{-3}$$

$R_4 = 8\,\Omega$, $L_4 = 19.09$ mH

$\bar{V} = 230\angle 0$ V at $f = 50$ Hz

$Z_1 = R_1 + j\omega L_1 = 10 + j\,2\pi \times 50 \times 31.8 \times 10^{-3}$

$= 10 + j10\ \Omega$

Compute $\bar{I}$, Powers of the source, pf

P, Q, S, pf

Solution: $Z_1 = R_1 + j\omega L_1 = 10 + j\,2\pi \times 50 \times 31.8 \times 10^{-3} = 10 + j10\ \Omega$

$Z_2 = R_2 + j\omega L_2 = 10 + j\,2\pi \times 50 \times 63.6 \times 10^{-3} = 10 + j20\ \Omega$

$Z_3 = R_3 - j/\omega C_3 = 3 - \dfrac{j}{2\pi \times 50 \times 795 \times 10^{-6}} = 3 - j4\ \Omega$

$Z_4 = R_4 + j\omega L_4 = 8 + j\,2\pi \times 50 \times 19.09 \times 10^{-3} = 8 + j6\ \Omega$

$$Z_{234} = \frac{1}{\left(\frac{1}{Z_2} + \frac{1}{Z_3} + \frac{1}{Z_4}\right)} = \frac{1}{Y_1 + Y_2 + Y_3} = \frac{1}{Y}$$

$= 4.09 - j1.24\ \Omega$

$Z_t = Z_1 + Z_{234} = 10 + j10 + 4.09 - j1.24$

$= 14.09 - j8.76 = 16.2\angle -31.85^\circ\ \Omega$

$$\bar{I} = \frac{\bar{V}}{Z} = \frac{230\angle 0}{16.2\angle -31.85^\circ} = 14.19\angle 31.85^\circ\ A$$

$\bar{S} = \bar{V}\,\bar{I}^* = 230\angle 0 \times 14.19\angle 31.85^\circ$

$= 3263.7\angle 31.85^\circ = 2772.3 + j1722.3$

$= P + jQ$

$P = 2772.3$ W

$Q = 1722.3$ VAr

$S = 3263.7$ VA

$pf = \cos(31.85^\circ) = 0.85$ (lag)

$\bar{V}_1 = \bar{V} - \bar{I}\,Z_1 = 230\angle 0 - 14.19\angle -31.85^\circ \times (10 + j10)$

$= 59.04\angle -48.7^\circ$ V

$$\bar{I}_2 = \frac{\bar{V}_1}{Z_2} = \frac{59.04\angle -48.7^\circ}{15 + j20} = 2.36\angle -101.8^\circ\ A$$

$$\bar{I}_3 = \frac{V_1}{Z_3} = \frac{59.04\angle -48.7^\circ}{3 - j4} = 11.85\angle 43.7^\circ\ A$$

$$\bar{I}_4 = \frac{V_1}{Z_4} = \frac{59.04\angle -48.7^\circ}{8 + j6} = 5.92\angle -85.62^\circ\ A$$

$\bar{I} = \bar{I}_2 + \bar{I}_3 + \bar{I}_4 = 14.19\angle -31.85^\circ$